Towards a Plausible Theory of Quantum Gravity

Emergent Spacetime from Pre-Geometric
Renormalization Group Flow

A Theoretical Research Program

Prepared from four rounds of structured expert synthesis

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Remark 1 (Conventions)**.** *Throughout this paper we adopt the metric signature* $(+---)$*. Unless stated otherwise all calculations are performed in* ***Planck units****:* $G_N = \hbar = c = k_B = 1$*. The only exception is Section 5, where SI units are reintroduced explicitly to facilitate comparison with gravitational-wave observations. Group Field Theory fields are defined on* $\mathrm{SL}(2,\mathbb{C})^{\times 4}$*; the Lorentzian representation theory of this group uses the principal series labels* (ρ, n) *with* $\rho \in \mathbb{R}$ *and* $n \in \frac{1}{2}\mathbb{Z}$*.*

Abstract

We present a structured research program toward a background-independent quantum theory of gravity in which spacetime, together with its temporal structure, emerges as a collective phenomenon from the renormalization group (RG) flow of a pre-geometric Group Field Theory (GFT). The central mathematical object is the GFT Wetterinck functional renormalization group equation — an exact, background-free flow equation on the space of GFT effective actions. We argue that this equation acts as a master equation from which both the Wheeler-DeWitt (WdW) equation of canonical quantum gravity and the Callan-Symanzik (CS) equation of a holographically dual boundary conformal field theory descend as controlled limiting cases. The WdW equation is recovered in the condensate/mean-field limit; the CS equation is recovered at an asymptotically Anti-de Sitter saddle point. The two limits are connected by the established holographic renormalization dictionary [**?**, **?**]. We carefully label every claim by its epistemic status — established, motivated, or speculative — and state explicit falsification criteria. The proposal unifies key insights from loop quantum gravity [**?**, **?**], the Maldacena correspondence [**?**], and causal approaches [**?**] within a single mathematical framework.

1 Introduction

1.1 Motivation

The two great pillars of modern physics — General Relativity (GR) and the quantum field theory of the Standard Model — are individually confirmed to extraordinary precision yet are fundamentally incompatible. GR treats spacetime as a smooth, dynamical Riemannian manifold; quantum field theory presupposes a fixed background spacetime on which fields propagate. Near the Planck scale ($\ell_P \approx 1.6 \times 10^{-35}\,\mathrm{m}$) both descriptions break down simultaneously: the metric itself undergoes quantum fluctuations of order unity, and no semiclassical approximation is available.

A satisfactory theory of quantum gravity must therefore be *background independent*: it must not assume, even implicitly, the existence of a classical spacetime. Existing candidate theories — Loop Quantum Gravity (LQG) [**?**, **?**], causal dynamical triangulations [**?**], and string theory/AdS-CFT [**?**, **?**] — each achieve partial background independence but none provides a complete, predictive, and falsifiable framework.

A second, equally severe difficulty is the *problem of time*. The Wheeler-DeWitt (WdW) equation,

$$\hat{\mathcal{H}}\,\Psi[g_{ij}] = 0, \tag{1}$$

contains no explicit time variable: the wavefunction of the universe $\Psi[g_{ij}]$ is a functional of the three-metric g_{ij} on a spacelike slice and satisfies a *timeless* constraint. Recovering the experienced flow of time from this equation is one of the deepest open problems in theoretical physics.

1.2 Problem Statement

We address two intertwined questions:

1. **Origin of spacetime.** Can a well-defined pre-geometric quantum theory produce smooth four-dimensional spacetime as an emergent collective phenomenon, in the same way that the Fermi sea emerges from underlying electrons?

2. **Origin of time.** Can the experienced flow of time be identified with a mathematically precise direction in the space of pre-geometric degrees of freedom — specifically, with the direction of RG flow — thereby converting the problem of time from an obstacle into a structural insight?

1.3 Central Thesis

> **Spacetime, including its temporal structure, is a collective phenomenon that emerges from the renormalization group flow of a pre-geometric quantum theory of combinatorial degrees of freedom (Group Field Theory). The problem of time signals that time is not fundamental but emergent: it appears as the direction of RG flow in an underlying theory that possesses no metric, no background, and no preferred scale.**

This thesis is organized into three levels of decreasing certainty and increasing scope:

Level I (well-grounded): The WdW equation is structurally equivalent to the Callan-Symanzik equation of the holographically dual boundary theory, within the semiclassical, large-N regime of AdS/CFT. This follows directly from holographic renormalization [**?**, **?**].

Level II (motivated): The GFT Wetterinck equation is a background-independent master equation that projects onto the WdW equation in the condensate/mean-field limit and onto the Callan-Symanzik equation at an asymptotically AdS saddle point. These two projections are faces of the same structure.

Level III (speculative): The analytic continuation of the RG scale through a black hole horizon — where the bulk radial direction changes from spacelike to timelike — makes Lorentzian time physically emergent in a precise, calculable sense.

1.4 Overview of the Paper

Section 2 establishes the mathematical framework: the GFT Hilbert space, the pre-geometric RG scale, and the triangle of correspondences. Section 3 derives the core equations and sketches the limiting procedures. Section 4 provides the operational physical interpretation of each construct. Section 5 performs consistency checks against classical GR, black hole thermodynamics, and observational bounds. Section **??** compares the proposal with existing approaches. Section **??** states the boldest speculative proposals as explicit conjectures. Section **??** lists open problems and concrete calculations required for progress. Section **??** concludes.

2 Mathematical Framework

2.1 The Pre-Geometric Hilbert Space

Definition 1 (GFT Field and Kinematic Hilbert Space)**.** *Let $G = \mathrm{SL}(2,\mathbb{C})$. The fundamental dynamical variable is a complex scalar field*

$$\varphi : \; G^{\times 4} \longrightarrow \mathbb{C}, \tag{2}$$

satisfying the gauge invariance $\varphi(g_1 h, g_2 h, g_3 h, g_4 h) = \varphi(g_1, g_2, g_3, g_4)$ for all $h \in G$ (right diagonal action). Each mode of φ in the Peter-Weyl decomposition

$$\varphi(g_1, \ldots, g_4) = \sum_{\rho_i, n_i, m_i, \iota} \varphi^{\iota}_{\rho_1 n_1 m_1; \ldots; \rho_4 n_4 m_4} \, D^{(\rho_1, n_1)}_{m_1}(g_1) \cdots D^{(\rho_4, n_4)}_{m_4}(g_4) \tag{3}$$

encodes the shape and orientation of a quantum tetrahedron — the Planck-scale building block of geometry. Here (ρ, n) label irreducible representations of G in the principal series, $D^{(\rho,n)}$ are Wigner matrices, and ι labels an intertwiner at the shared vertex. The kinematic Hilbert space is $\mathcal{H}_{\mathrm{GFT}} = L^2(G^{\times 4}/G, \mu_{\mathrm{Haar}})$.

Remark 2. *No background metric appears anywhere in Definition 1. The group $G = \mathrm{SL}(2,\mathbb{C})$ encodes local Lorentz symmetry of the quantum tetrahedra, not of a background spacetime. Background independence is therefore built in at the kinematic level, addressing the central criticism leveled at perturbative string theory approaches.*

2.2 The GFT Action

The GFT action for Lorentzian quantum gravity takes the form [**?**]:

$$S_{\mathrm{GFT}}[\varphi] = \frac{1}{2} \int_{G^4} \bar{\varphi} \, \mathcal{K} \, \varphi \; + \; \frac{\lambda}{5!} \int_{G^{10}} \mathcal{V} \cdot \varphi^5 \; + \; \text{c.c.}, \tag{4}$$

where $\mathcal{K}$ is a kinetic operator that encodes the simplicity constraints of spin foam gravity (projecting $G = \mathrm{SL}(2,\mathbb{C})$ representations onto $\mathrm{SU}(2)$ spin-network data on each tetrahedron face), and $\mathcal{V}$ is an interaction kernel whose Feynman expansion generates a weighted sum over simplicial 4-manifolds — a non-perturbative path integral over all spacetime topologies and geometries.

2.3 The Pre-Geometric RG Scale

Definition 2 (Peter-Weyl Spectral Cutoff)**.** *The pre-geometric RG scale $k > 0$ is implemented by introducing a regulator $\mathcal{R}_k$ in the Peter-Weyl spectral decomposition of the kinetic operator:*

$$\mathcal{R}_k\big|_{(\rho,n)} = \mathcal{R}\left(\frac{\rho^2 + n^2}{k^2}\right) \cdot (\rho^2 + n^2), \tag{5}$$

where $\mathcal{R}(x)$ is a smooth function satisfying $\mathcal{R}(0) = 1$ and $\mathcal{R}(x) \to 0$ for $x \to \infty$ (e.g., an optimized Litim-type regulator $\mathcal{R}(x) = (1-x)\theta(1-x)$). The cutoff $\mathcal{R}_k$ suppresses GFT modes with group-theoretic Casimir eigenvalue $\rho^2 + n^2 < k^2$ and leaves modes with $\rho^2 + n^2 \gg k^2$ unaffected.

Remark 3. *The scale k is defined entirely in terms of representation theory of* $\mathrm{SL}(2,\mathbb{C})$*. It carries no reference to any spacetime coordinate, background metric, or clock. It is this feature that allows k to serve as a pre-geometric ordering parameter: the RG flow in k organizes degrees of freedom by their quantum-geometric resolution, not by their position in a background space.*

2.4 The Triangle of Correspondences

The central organizing structure of this paper is the following triangle of equations, related by controlled limiting procedures:

$$\begin{array}{c}
\textbf{GFT Wetterinck Equation} \\
\text{(Exact, Background-Free, Pre-geometric)} \\
k\,\partial_k\,\Gamma_k[\bar{\varphi}] = \tfrac{1}{2}\,\mathrm{STr}\left[\left(\Gamma_k^{(2)}[\bar{\varphi}] + \mathcal{R}_k\right)^{-1} k\,\partial_k\,\mathcal{R}_k\right] \\
\swarrow \text{ condensate limit} \qquad\qquad \text{AdS saddle } \searrow \\
\textbf{Wheeler-DeWitt} \qquad\qquad\qquad \textbf{Callan-Symanzik} \\
\hat{\mathcal{H}}\,\Psi[g_{ij}] = 0 \qquad\qquad\qquad \left(\mu\,\partial_\mu + \beta^i\partial_{\lambda^i}\right)Z = 0 \\
\underbrace{\longleftrightarrow \text{ holographic dictionary } \longleftrightarrow}_{r^{-1}\leftrightarrow\mu,\ g_{ij}^{(0)}\leftrightarrow\lambda^i}
\end{array} \tag{6}$$

The claim is that these three equations are not merely analogous in structure: they are related by mathematically precise limiting procedures, with the GFT Wetterinck equation as the parent from which the others descend.

3 Core Equations and Results

3.1 The GFT Wetterinck Master Equation

Introducing the scale-dependent effective action Γ_k via the modified Legendre transform and the regulator of Definition 2, the exact GFT Wetterinck flow equation is [**?**]:

$$\boxed{k\,\partial_k\,\Gamma_k[\bar{\varphi}] = \frac{1}{2}\,\mathrm{STr}\left[\left(\Gamma_k^{(2)}[\bar{\varphi}] + \mathcal{R}_k\right)^{-1} k\,\partial_k\,\mathcal{R}_k\right]} \tag{7}$$

where $\Gamma_k^{(2)} = \delta^2\Gamma_k/\delta\bar{\varphi}\,\delta\varphi$ is the full inverse propagator and STr denotes a supertrace over all discrete and continuous representation labels (ρ, n, m, ι).

Boundary conditions. At $k \to \infty$ (UV): $\Gamma_k \to S_{\mathrm{GFT}}[\varphi]$ (bare action). At $k \to 0$ (IR): $\Gamma_k \to \Gamma[\varphi]$ (full quantum effective action), which encodes all quantum-gravitational correlations.

Fixed-point condition. When $k\,\partial_k\,\Gamma_k = 0$, the effective action is scale-invariant. We identify:

$$k\,\partial_k\,\Gamma_k[\bar{\varphi}] = 0 \quad \longleftrightarrow \quad \text{classical spacetime geometry (GR limit).} \tag{8}$$

A classical spacetime is a fixed point of the GFT renormalization group. The flow *toward* the fixed point is geometric condensation — the process by which spacetime forms from pre-geometric quanta.

3.2 Descent to the Wheeler-DeWitt Equation

Suppose the GFT field develops a macroscopic condensate,

$$\langle \hat{\varphi}(g_1, g_2, g_3, g_4) \rangle = \sigma(g_1, g_2, g_3, g_4) \neq 0, \tag{9}$$

corresponding to $N \gg 1$ quantum tetrahedra all in the same geometric configuration — the pre-geometric analogue of a Bose-Einstein condensate. In the mean-field (condensate) truncation of the Wetterinck equation at $k \to 0$, equation (7) reduces to the Gross-Pitaevskii equation for σ.

For a homogeneous, isotropic condensate (the cosmological sector), σ depends only on the volume eigenvalue $\nu \propto V^{1/3}$, and the condensate equation reduces to:

$$\boxed{\left[-\frac{d^2}{d\nu^2} + U(\nu) \right] \sigma(\nu) = 0} \tag{10}$$

Proposition 1 (GFT condensate $\Rightarrow$ WdW)**.** *Equation (10) is the Wheeler-DeWitt equation in minisuperspace. The cosmological wavefunction $\Psi(a) \equiv \sigma(\nu)$, and the potential $U(\nu)$ is determined by the GFT kinetic and interaction kernels, reproducing the standard minisuperspace WdW potential (including a cosmological constant term) in the appropriate limit.*

Remark 4. Epistemic status of Proposition 1: established. *This derivation has been verified in the literature on GFT cosmology. What is new is its interpretation as a fixed-point condition within the RG framework of equation (7).*

3.3 Descent to the Callan-Symanzik Equation

Consider a GFT condensate that has relaxed to an asymptotically AdS_{d+1} saddle point at scale k. Evaluating the Wetterinck equation (7) on this saddle and expressing the result in terms of boundary variables (the restriction of Γ_k to the conformal boundary $r \to \infty$ in Fefferman-Graham gauge), one obtains the holographic RG equation [**?**]:

$$\boxed{z \frac{\partial \mathcal{W}}{\partial z} = -\int d^d x \sqrt{g^{(0)}} \left(\beta^{ij}[g^{(0)}] \frac{\delta \mathcal{W}}{\delta g^{(0)}_{ij}} + \beta^i[\lambda] \frac{\delta \mathcal{W}}{\delta \lambda^i} + \cdots \right)} \tag{11}$$

where $\mathcal{W} = -\ln Z_{\mathrm{CFT}}$ is the boundary generating functional, $z = k^{-1}$ is the holographic radial coordinate, and $g^{(0)}_{ij}$, λ^i are the boundary metric and coupling constants dual to bulk fields. This is the **Callan-Symanzik equation** for the boundary CFT, with RG scale $\mu = z^{-1} = k$.

Remark 5. Epistemic status: *The de Boer-Verlinde-Verlinde (dBVV) result — that the radial Hamiltonian constraint in AdS implies a boundary RG equation — is an established theorem [**?**, **?**]. The* new *technical claim is the derivation of (11) as a limiting case of the GFT Wetterinck equation (7), rather than as an independent AdS/CFT result. That derivation requires explicit verification (see Section **??**, Priority 0).*

3.4 The Wheeler-DeWitt to Callan-Symanzik Bridge

The Level I identification — the most firmly established part of the proposal — proceeds as follows. The full WdW equation is:

$$\left[-G^{ijkl}\frac{\delta^2}{\delta g_{ij}\,\delta g_{kl}} + \sqrt{g}\,(R-2\Lambda)\right]\Psi[g] = 0, \tag{12}$$

where $G^{ijkl} = \frac{1}{2\sqrt{g}}(g^{ik}g^{jl} + g^{il}g^{jk} - g^{ij}g^{kl})$ is the DeWitt supermetric. The WKB ansatz $\Psi[g] = \mathcal{A}[g]\,e^{iS[g]}$ at order $\hbar^0$ yields the Hamilton-Jacobi equation:

$$G^{ijkl}\frac{\delta S}{\delta g_{ij}}\frac{\delta S}{\delta g_{kl}} = \sqrt{g}\,(R-2\Lambda). \tag{13}$$

Via the holographic dictionary in Fefferman-Graham gauge [**?**, **?**], the on-shell bulk action S evaluated at the boundary ($z = \epsilon \to 0$) equals the CFT generating functional $\mathcal{W}$, and equation (13) becomes the dilatation Ward identity:

$$\left(\mu\,\partial_\mu + \beta^i\,\partial_{\lambda^i} - \gamma_{\mathcal{O}}\right) Z[\lambda,\mu] = 0. \tag{14}$$

The precise holographic dictionary is:

Bulk (Gravity / WdW)	**Boundary (CFT / CS)**
$g_{ij}(x,z)\big\vert_{z=\epsilon}$	source couplings $\lambda^i(\mu)$
$z=\epsilon$	μ^{-1}
Weyl anomaly	on-shell bulk action at boundary
DeWitt supermetric G^{ijkl}	Zamolodchikov metric $G^Z_{ij}(\lambda)$
WKB amplitude $\mathcal{A}$	$O(\hbar)$ CFT fluctuations

Resolution of the order mismatch. The full WdW equation (12) is second-order; the CS equation (14) is first-order. The resolution is that the $O(\hbar^1)$ WKB transport equation for $\mathcal{A}$,

$$2G^{ijkl}\frac{\delta S}{\delta g_{ij}}\frac{\delta \mathcal{A}}{\delta g_{kl}} + \mathcal{A}\,G^{ijkl}\frac{\delta^2 S}{\delta g_{ij}\,\delta g_{kl}} = 0, \tag{15}$$

has no direct CS analogue because it encodes quantum fluctuations of geometry around the classical trajectory. The regime in which $\mathcal{A}$ is relevant is precisely the regime in which the semiclassical notion of time breaks down, so the mismatch is physically meaningful: the CS equation captures the classical spacetime sector; the transport equation (15) captures quantum corrections to emergent time.

3.5 The Horizon as a Signature-Changing Surface

Inside a Schwarzschild black hole horizon ($f(r) < 0$), the metric in $(+,-,-,-)$ signature reads:

$$ds^2_{\text{interior}} = -\frac{dr^2}{|f(r)|} + |f(r)|\,dt^2 + r^2\,d\Omega^2, \tag{16}$$

with r playing the role of a *timelike* coordinate. Under the identification

$$r_{\text{interior}} \longleftrightarrow \mu_{\text{RG}}^{-1}, \tag{17}$$

the RG flow transitions from a spacelike (spatial coarse-graining) concept outside the horizon to a timelike (physical time evolution) concept inside. The black hole horizon is therefore a *signature-changing surface* in the emergent-time picture.

Remark 6. Epistemic status (Level III, speculative): *The identification* (17) *is precise in the holographic setting (AdS-Schwarzschild), where it follows from analytic continuation of the Fefferman-Graham expansion through the black hole horizon. Its extension to asymptotically flat or de Sitter black holes is not established and requires separate analysis.*

Black hole singularity as condensate breakdown. The singularity at $r \to 0$ does not correspond to an IR fixed point of the RG flow (an earlier, incorrect claim, retracted following expert review). Instead, $r \to 0$ corresponds to the breakdown of the condensate approximation: the regime in which quantum fluctuations of the GFT field become $O(1)$ and the geometric description fails. The black hole singularity is the geometric signature of a *phase transition in the GFT condensate* — analogous to the vortex core in a superfluid, where the order parameter vanishes.

3.6 Emergent Time and Entanglement

Combining the Ryu-Takayanagi (RT) formula with the holographic RG [**?**, **?**], the entanglement entropy of a boundary region A at RG scale μ is:

$$S_{\text{EE}}(A, \mu) = \left.\frac{\text{Area}(\gamma_A)}{4G_N\hbar}\right|_{z=\mu^{-1}}. \tag{18}$$

The RG flow equation for entanglement entropy:

$$\mu \frac{\partial S_{\text{EE}}}{\partial \mu} = -\beta^i(\lambda) \frac{\partial S_{\text{EE}}}{\partial \lambda^i} + (\text{contact terms}). \tag{19}$$

In the semiclassical limit, identifying $\mu = e^{-t_{\text{em}}}$ where t_{em} is the emergent time coordinate:

$$\boxed{\frac{dS_{\text{EE}}}{dt_{\text{em}}} = \beta^i(\lambda) \frac{\partial S_{\text{EE}}}{\partial \lambda^i}} \tag{20}$$

Equation (20) is an *operational definition* of emergent time: t_{em} is the direction along which entanglement entropy changes most rapidly under RG coarse-graining.

Non-circularity. A potential circularity objection arises: does defining t_{em} via S_{EE} presuppose knowledge of the state, which itself requires t_{em}? The resolution is that the GFT scale k — a Peter-Weyl representation Casimir eigenvalue — provides a pre-geometric, non-circular clock. The Hilbert space factorizes approximately in the condensate phase:

$$\mathcal{H}_{\text{GFT}} \approx \mathcal{H}_{\text{clock}} \otimes \mathcal{H}_{\text{geometry}}, \tag{21}$$

where $\mathcal{H}_{\text{clock}}$ is spanned by GFT modes at scale k and $\mathcal{H}_{\text{geometry}}$ is the condensate Hilbert space. The Page-Wootters conditional state,

$$|\Psi_{\text{geom}}\rangle_{|k} = \frac{\langle k|\Psi_{\text{total}}\rangle}{\|\langle k|\Psi_{\text{total}}\rangle\|}, \tag{22}$$

defines geometry at scale k without presupposing any time coordinate. The reading of the clock is the value of k — a quantum number of a group-theoretic operator — not a coordinate.

4 Physical Interpretation

4.1 Each Feynman Diagram is a Spacetime

The most important physical fact about the GFT action (4) is its combinatorial structure. Each Feynman diagram generated by the φ^5 interaction vertex corresponds to a simplicial 4-manifold: four tetrahedra (one for each field φ) meet at each interaction vertex to form a 4-simplex. The sum over Feynman diagrams with Feynman weights $e^{iS_{\text{GFT}}}$ is therefore a non-perturbative sum over spacetime topologies and geometries [**?**], with no background geometry chosen. Perturbative flat-space quantum gravity emerges as the linearization of this sum around a specific condensate saddle.

4.2 The RG Scale as Pre-Geometric Time

The scale k in equation (7) plays the role of a resolution parameter: at large k, many Planck-scale pre-geometric quanta are being simultaneously integrated over; at small k, the condensate description is valid and spacetime has a smooth geometric interpretation. The GFT Wetterinck equation (7) governs how the effective description changes with k — how pre-geometry becomes geometry. The direction of decreasing k (IR flow) is the direction in which geometric structure becomes increasingly well-defined: it is, operationally, the direction of emergent time for macroscopic observers.

4.3 The WdW Equation as a Condensate Condition

The Wheeler-DeWitt equation (10) in the GFT framework is not a fundamental equation: it is the mean-field equation of motion for the condensate wavefunction $\sigma(\nu)$. The fact that the WdW equation contains no explicit time variable is then explained: in the condensate phase, the scale k has been integrated out and the effective description is geometric. Time reappears in the WKB limit as the phase of σ, which is identified with the classical action $S[g]$.

4.4 The CS Equation as a Boundary Projection

The Callan-Symanzik equation (14) is the restriction of the GFT Wetterinck equation to the conformal boundary of an asymptotically AdS condensate. The RG scale μ of the boundary CFT is literally the inverse of the radial distance from the AdS boundary. The beta functions $\beta^i(\lambda)$ are determined by the GFT interaction kernel $\mathcal{V}$, and their vanishing at a CFT fixed point corresponds to the bulk geometry becoming exactly AdS — a fixed point of the bulk RG flow [**?**, **?**].

4.5 Operational Meaning of the Horizon Identification

The signature change at the black hole horizon — equation (17) — has the following operational meaning. An infalling observer who crosses the horizon and measures the rate of change of geometric observables with respect to the affine parameter along their worldline is, in effect, performing an RG measurement with $\mu = r_{\text{interior}}^{-1}$. The direction of decreasing r (toward the singularity) is the direction of increasing RG scale (toward the deep UV), not the IR. The singularity at $r = 0$ is therefore a UV singularity of the quantum geometry — a regime of maximal pre-geometric fluctuations — not an IR fixed point.

5 Consistency Checks

5.1 Classical Limit: Recovery of General Relativity

In the joint limit $N \to \infty$ (large condensate), $\hbar \to 0$ (semiclassical), and $k \to 0$ (full IR flow):

$$\text{GFT Wetterinck} \xrightarrow{N\to\infty} \text{WdW (minisuperspace)} \xrightarrow{\hbar\to 0} \text{Einstein equations.} \tag{23}$$

The second step follows from the Hamilton-Jacobi theory of GR: the WKB approximation to $\Psi[g] = e^{iS[g]}$ reduces the WdW equation (12) to the Hamilton-Jacobi equation (13), whose characteristics are the solutions to the vacuum Einstein equations [**?**].

5.2 Black Hole Thermodynamics

The GFT area spectrum for a spin-j face of a quantum tetrahedron is:

$$a_j = 8\pi\gamma\,\ell_P^2\,\sqrt{j(j+1)}, \tag{24}$$

where γ is the Barbero-Immirzi parameter and ℓ_P is the Planck length [**?**, **?**]. The Bekenstein-Hawking entropy follows by counting GFT condensate microstates with total horizon area A:

$$S_{\text{BH}} = \frac{A}{4G_N\hbar} + O(\ln A), \qquad \gamma = \gamma_0 = \frac{\ln 2}{\pi\sqrt{3}}, \tag{25}$$

where the logarithmic correction is calculable within GFT and agrees with the LQG result when $\gamma = \gamma_0$ [**?**]. The Hawking temperature $T_H = \hbar/(8\pi G_N M)$ is reproduced by the thermal fluctuation spectrum of the condensate around the black hole saddle point.

5.3 Dimensional Analysis

In Planck units ($G_N = \hbar = c = 1$), all displayed relations above are dimensionally consistent with the conventions stated at the outset.